Predicting Groundwater Levels in Southern New Mexico using Deep Learning

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1 Introduction

The western United States is increasingly facing severe water scarcity that threatens agricultural production, economic growth, and ecological stability. The decline in natural replenishment driven by climate change strains limited water supplies. In many places, this is exacerbated by rising consumption rates. In New Mexico, groundwater extraction has significantly outpaced aquifer recharge, leading to significant declines in groundwater levels in many areas[1]. Increasing aridification is expected to cause large decreases in surface water availability in the coming years, increasing demand for groundwater. This has been especially evident in the Rio Grande watershed this year. The current mega-drought and lack of snow pack in southern Colorado and northern New Mexico, combined with large amounts of groundwater extraction led to the Rio Grande being entirely dry in Albuquerque for approximately three months[3]. In southern New Mexico, Elephant Butte Reservoir reached its lowest level since 1971, and in July was at only 1.4 percent of capacity[5]. These hydrological stresses have contributed not only to interstate legal disputes, especially regarding compliance with the Rio Grande Compact, but also impacted agriculture and other economic areas, such as a chile harvest that is anticipated to be between twenty and fifty percent smaller than expected. [4].

Physically based hydrological models that simulate interactions between precipitation, temperature, evaporation, surface water, and groundwater are a common approach to quantifying water resources. For example, the Rio Grande Transboundary Hydrologic Integrated Model (RGTHIM) was developed to represent the influence of precipitation on conjunctive use and management of water resources in Southern New Mexico. This model is highly demanding of computational resources, which makes it difficult to use for evaluating parameter uncertainty and various management policies. Data-driven approaches may be able to support more rapid assessment and continuous refinement of water-management strategies. This study explores the use of deep learning techniques to predict groundwater levels in southern New Mexico using long-term datasets of precipitation, temperature, streamflow, and groundwater levels, with the goal of evaluating whether a deep learning model can accurately reproduce hydrological patterns and deliver predictions suitable for guiding water management strategies.

2 Dataset

Precipitation and temperature records have been obtained from the National Oceanic and Atmospheric Administration (NOAA) station located just north of Las Cruces, New Mexico (red point, Figure 1) [6]. Streamflow data and groundwater level data have been obtained from the United States Geological Survey (USGS), for a stream gauge on the Rio Grande below Elephant Butte Dam (yellow point, Figure 1)[9] and twenty-three wells (blue points, Figure 1)[8]. The precipitation and temperature dataset contains approximately 24,000 daily observations spanning 67 years, each including the quantity of precipitation measured and the maximum and minimum temperature for the day. Of these, fewer than 59 values are missing. Streamflow data consisted of 20,000 observations across 55 years, and groundwater well sampling frequency and temporal coverage varied widely.

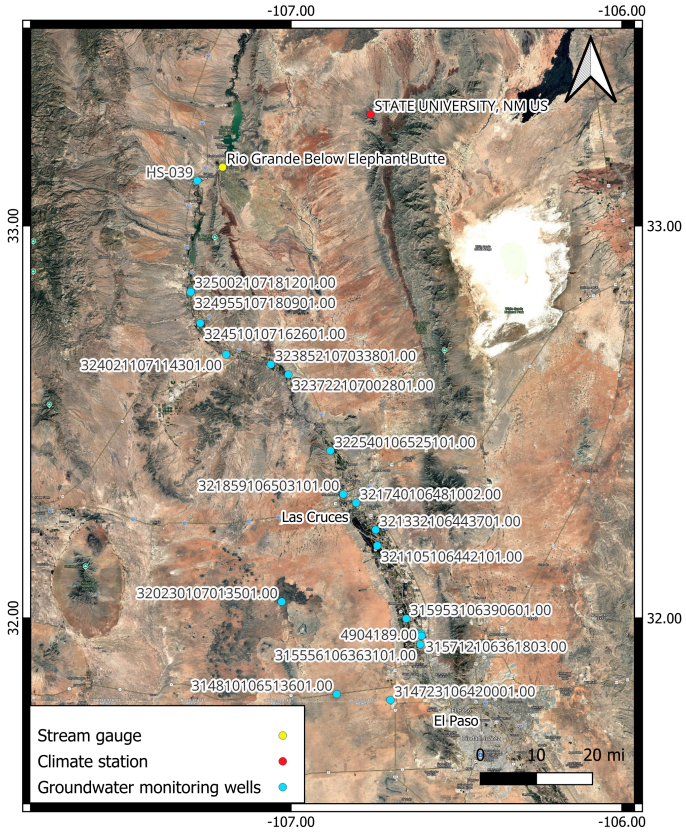


Figure 1: Location of Climate, Streamflow, and Groundwater Level stations

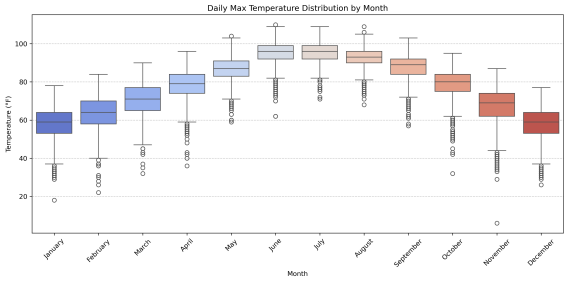
2.1 Missing Values

Within the precipitation dataset, 1 precipitation value was missing from 1978, 1 value was missing from 2026, and 28 were missing from November of 1997. Missing values for 1978 and 2026 were imputed from the average of the previous and subsequent days. For November of 1997, the mean daily precipitation of the months of November from 1995 through 1999 was used as the imputation value. The maximum temperature data was missing 46 values. For values missing on fewer than 3 consecutive days, the temperature was again imputed from the average of the previous and subsequent days, while for the 28 missing November 1997 values, the mean of Novembers from 1995 through 1999 was again used. For May of 2017, where 3 consecutive days were missing, the average of the day preceding and the day following the missing block of days was used. The minimum temperature was missing 59 values. Again, November of 1997 was missing, and was imputed from Novembers as above. Streamflow and groundwater data had no missing values.

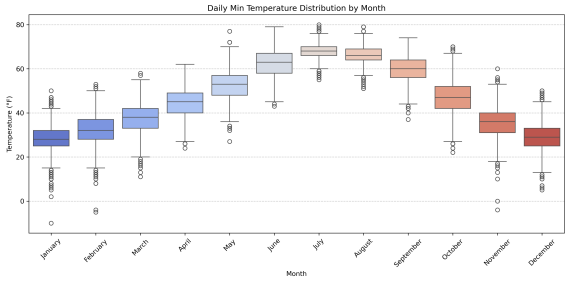
2.2 Statistical Distributions

The precipitation records span the years 1959 through 2026. Throughout the years, the minimum precipitation recorded was, unsurprisingly, 0, and the maximum was 3.34 inches in a single day, with a daily mean of 0.025 inches. Not surprisingly, the precipitation data shows significant positive skew, reflecting New Mexico's arid climate, with largely monsoon-driven outliers. Similarly, overall precipitation data showed excessive kurtosis, indicating that extreme values are common. As expected, while this trend continued when considering distribution for each month, e.g. all the values in any December, it was much smaller than when considering distribution over each year. Additionally, precipitation data shows very little correlation between either maximum or minimum temperatures and precipitation levels. Temperature data was nearly symmetric with a standard deviation of approximately 22 F, consistent with a dataset spanning many years and exhibiting broad variability, but few sharp peaks or heavy tails. Streamflow data spans the years 1971 through 2026, and exhibits moderate right skew

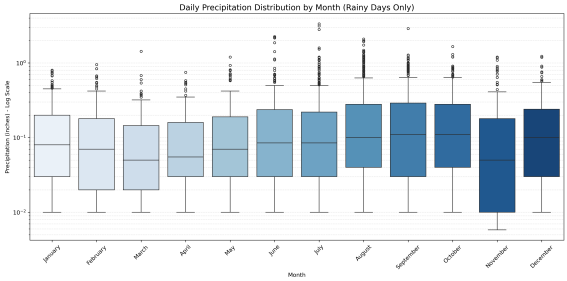
(0.75) and large variability (SD ≈ 913 cubic feet per second). Overall kurtosis was very slight and negative, suggesting that in general the use of Elephant Butte as a reservoir is sufficient to keep downstream flow on a normal distribution. Monthly water flow is generally highest in the summer, and lowest in the winter, in keeping with the general use of reservoir water for agriculture. Additionally, several months show significant numbers of outliers in streamflow rates, suggesting that those months are prone to additional upstream precipitation events and other impacts, where river flow may be unmeasurable in dry spells, or very high for storm related high-flows or large dam releases. Groundwater records spanned 1952 through 2026, with data available for a minimum of over fifteen years for the shortest running well data, to over 73 years for the well with the longest collection time. The mean length of time data was collected was 37.56 years. Water level distributions tended to have very small negative skew, and slight positive kurtosis, although Well 6 had much more negative skew and much higher positive kurtosis, which warrants additional consideration as a general outlier in distribution. By well, the maximum number of observations was over 14,000, and a mean number of observations per well of 3836. The Figure 2 shows monthly boxplots of Maximum and Minimum Temperature, Precipitation, and Streamflow, while the Figure 3 shows boxplots of all 23 wells, demonstrating that the distribution of groundwater levels is consistent with the elevation of the wells from north to south.



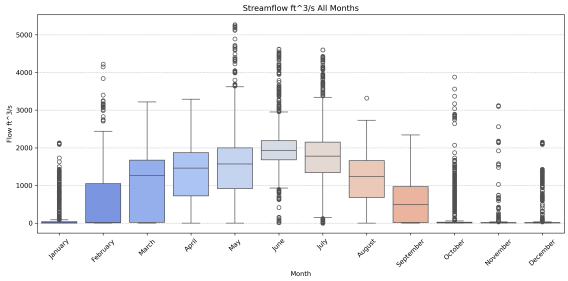
(a) Maximum temperature



(b) Minimum temperature



(c) Precipitation



(d) Streamflow

Figure 2: Monthly distributions of daily maximum temperature, minimum temperature, precipitation, and streamflow.

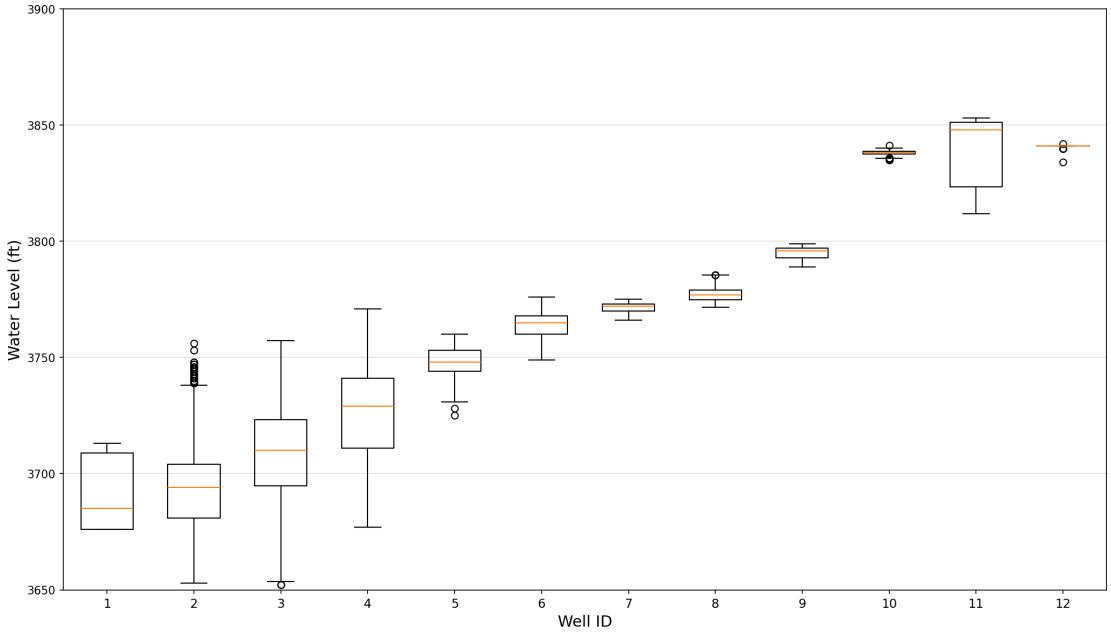
2.3 Data Pre-processing

To create a consistent monthly dataset suitable for machine learning models, the following pre-processing steps were applied:

1. Selection of a Common Modeling Period: March 1994 - December 2025 to maximize overlapping well coverage.

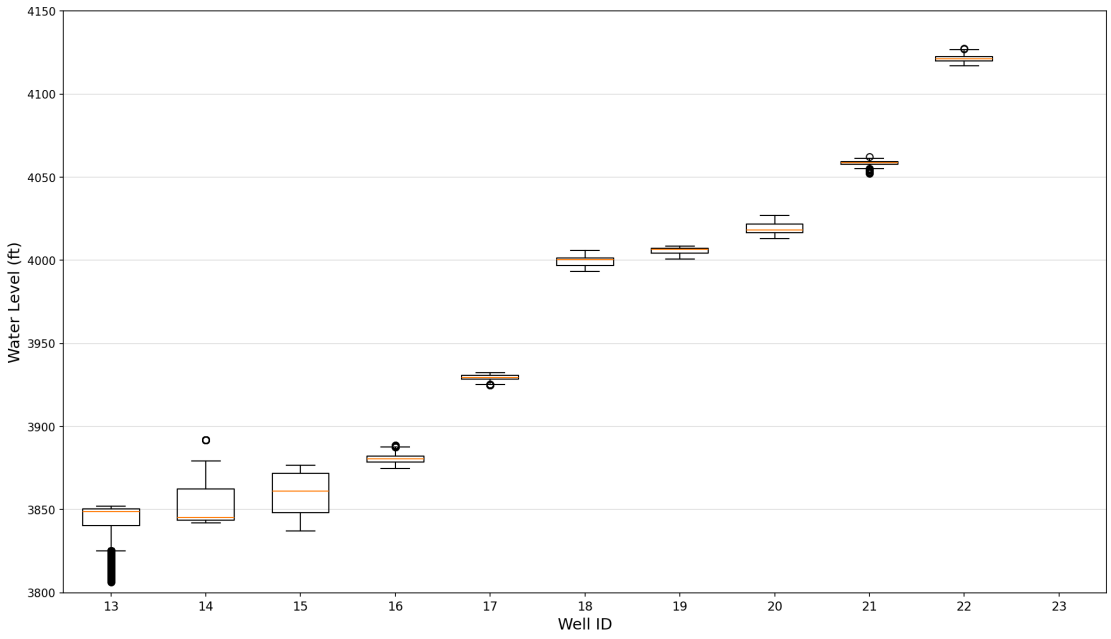
2. Monthly averages of daily data: Climactic variables have been aggregated to monthly averages; groundwater monthly values were computed from available measurements. This assumption is valid, as monthly-scale variations in groundwater levels are less significant than those of climactic variables.

3. Filtering Wells based on Data Availability: wells with more than 40% of their data were removed, reducing wells from 22 to 7.



ID	1	2	3	4	5	6	7	8	9	10	11	12
Well SiteNo	4904189	4904476	315712106361803	4904106	315556106363101	315901106355001	314723106420001	315953106390601	314810106513601	321105106442101	321332106443702	320230107013501

(a) Wells with the lowest mean water levels



ID	13	14	15	16	17	18	19	20	21	22	23
Well SiteNo	321332106443701	321740106481002	321740106481001	321859106503101	322540106525101	323722107002801	323733107011002	323930107041401	324021107114301	325002107181201	HS-039

(b) Wells with the highest mean water levels

Figure 3: Distribution of groundwater levels for the monitoring wells. Well IDs correspond to the ranking of wells by mean water level.

4. Interpolation of Remaining Missing Values: Among retained wells, some months still lacked measurements. To ensure continuity in the groundwater time series, Linear interpolation was applied to the missing groundwater months.

3 Proposed Methodology

To establish a baseline simple machine learning model, we will use an Xtreme Gradient Boost (XGBoost), which has been shown to perform well in predicting groundwater levels in arid regions([7]). We will compare this to a Recurrent Neural Network (RNN), most likely a Long Short-Term Memory (LSTM) model. RNNs have proven successful in regressive groundwater modeling [2], and LSTMs are well suited to regression problems involving temporal components. This will enable us to compare performance between a simple machine learning model and a deep learning model across several areas.

4 Evaluation

Models will be trained using a standard 70/30 training/test split, without shuffling the data in order to preserve seasonal and annual patterns. Because hydrologic variables often show strong temporal dependencies, we anticipate needing to time-lag some variables to provide the models with a means to learn seasonal cycles and long-term trends. For this comparison, we will calculate the mean squared error, the mean absolute error, and the R-squared score between every model, as well as the time required to run the model.

5 Expected Outcomes

We anticipate being able to predict groundwater levels with both models. However, we anticipate better performance from the LSTM, along with improved computational efficiency, measured in terms of time to perform computations and memory usage. We believe the most likely challenge will be a shortage of available groundwater data in the watershed of choice, due to differing sampling frequencies and periods across wells. Additionally, we are using a relatively small dataset, which may present challenges for effective training of the LSTM.

6 Timeline and milestones

- September 23: Create GitHub repository, upload all documents, datasets
- September 30: Finish data compilation, pre-processing, and initial exploration
- October 14: Complete XGBoost model
- October 28: Complete LSTM model
- November 6: Complete initial comparison of XGBoost, LSTM
- November 11: Complete any additional modifications required to the models after comparison, such as parameter tuning
- November 21: Complete written report
- December 2: Complete presentation

7 References

References

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[8] United States Geological Survey. National ground-water monitoring network.

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